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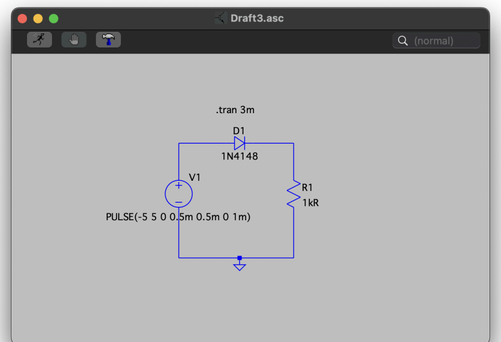
ECE 65

Pre-Lab 3-Diode iv Characteristics, Zener Diode, Clipper circuit

Experiment 1:

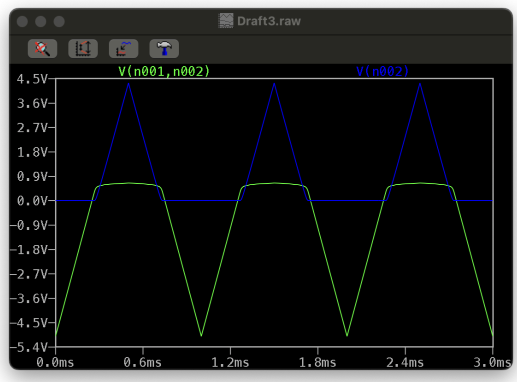
Simulation:

1)



←Figure 1

Figure 2 →



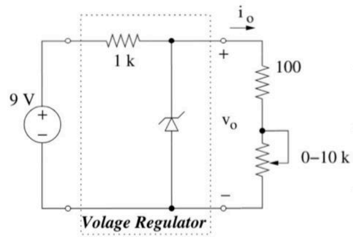
Created the schematic **(Figure1)** given in the lab into LTspice and plotted the graph **(Figure 2)** V_D = V(n001, n002) and V_R = V(n002)

2)

The forward-bias region is where V_D is greater than 0, but the diode is forward-biased not until around 0.6V.
The reverse bias region would be where V_D is less than 0V.

Experiment 2:

Circuit Analysis:



Zener Region

$V_o = -V_D = -5.6V$, but $i_D \leq 0$ so not possible

Diod on

$V_o = -V_D = -0.7V$, but $i_D \geq 0$ so not possible

Diode off

$V_o = -V_D$, $i_D = 0$, so $V_D = -i_o(1000\Omega) \Rightarrow V_o = 1000 i_o$

Maximum Load Current

Zener Region

$1000 i_1 + 5.6V = 9V \Rightarrow 1000 i = 3.4V \Rightarrow i_1 = 3.4mA$ $\quad i_o = \frac{5.6V}{100} = 56mA$ not possible $i_1 < i_o$

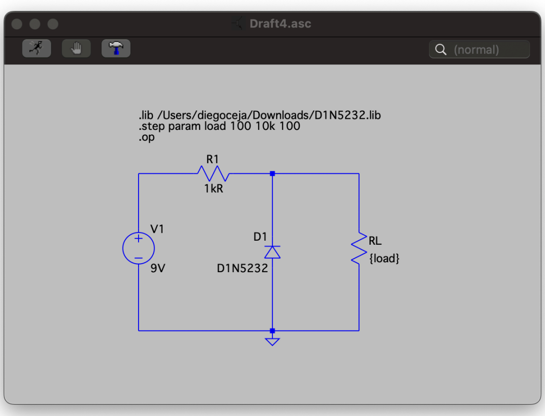
Diode on

$1000 i_1 = 9V + 0.7V \Rightarrow 1000 i = 9.7V \Rightarrow i_1 = 9.7mA$ $\quad i_o = \frac{-0.7V}{100} = -7mA$ not possible $i_1 > i_o$

Diode off $\qquad$ only possible state

$i_1 = i_o \Rightarrow 1000 i_o + 100 i_o = 9V \Rightarrow i_o = 8.18 mA$ $\quad i_o = \frac{V_o}{100} \Rightarrow V_o = (8.18)(100) = .818V \Rightarrow V_o = -0.818V$ so $i_o = 8.818 mA$

Simulation:



←Figure 3

Figure 4 →

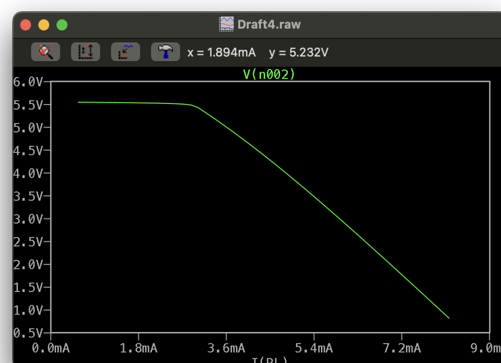


Figure 4 is where the output current is its maximum value, which is approximately 0.818V and the current through the V_1, R_1, and R_L, are all approximately 0.818mA, matching my results obtained through the circuit analysis.